Command sequence in “main_interfaz.f90”

This document contains the sequence of commands that “main_interfaz” performs. If you only are interested in the chemical part (ie. you solve transport separately) then you should only pay attention to the steps written in **bold**. The sequence is:

1. Create transient reactive transport class object: `my_RT_trans`
2. Create transient transport class object: `my_tpt_trans`
3. **Create chemistry class object:** `my_chem`
4. **Write directory of chemical databases** (including backslash or front slash depending on your OS)
5. **Write directory of problem to be solved** (including backslash or front slash depending on your OS)
6. **Write root of input and output files**
7. **Read chemistry:** `my_chem` **calls** `read_chemistry`
8. **Write name of file with concentrations after reactive mixing iteration**
9. **Choose if you want the code to compute the first iteration of conservative transport of components (**`opc_u_tilde`**=0) or if you want it to read the values from a file (**`opc_u_tilde`**=1)**
10. If opc_u_tilde=0:
    - Write file where you want to write the computed values
    - Choose if you want the code to compute mixing ratios (`tpt_opt`=0) or to read them (`tpt_opt`=1)
    - If tpt_opt=0:
        1. Choose type of mesh (1: 1D homogeneous, 2: 1D heterogeneous, 3: radial)
        2. Read transport data: `my_tpt_trans` calls subroutine `initialise_transport_1D_transient_RT`
        3. Allocate transport arrays: `my_tpt_trans` calls subroutines `allocate_arrays_PDE_1D` and `allocate_conc`
        4. Compute mixing ratios: `my_tpt_trans` calls subroutine `compute_mixing_ratios_Delta_t_homog`
        5. Set transport attribute in reactive transport object: `my_RT_trans` calls `set_transport_trans`
        6. Write integration method for chemical reactions and set its attribute: `my_RT_trans` calls `set_int_method_chem_reacts`
    - If tpt_opt =1:
        - Set transport attribute in reactive transport object: `my_RT_trans` calls `set_transport_trans`
        - Read time discretisation: `my_RT_trans` calls `read_time_discretisation`
        - Read transport data to apply WMA: `my_RT_trans` calls `read_transport_data_WMA`
   - Get initial time step: `time_discr` attribute of `my_tpt_trans` calls `get_Delta_t`
   - Compute first iteration of conservative transport: `my_chem` calls `compute_u_tilde_init`
   - Write computed values: `my_chem` calls `write_u_tilde_init`
11. **If opc_u_tilde=1:**
    - **Write name of file containing component concentrations after first iteration of conservative transport**
    - **Write initial time step**
    - **Specify if time step is constant (1) or not (0)**
12. **Get number of aqueous components:** `my_chem` **calls** `get_num_aq_comps`
13. **Start reactive mixing loop:**
    - **Compute reactive mixing iteration:** `my_chem` **calls** `interfaz_comps_arch`
    - **Decide if you want to continue iterating (1) or not (0)**
    - **If you want to continue and time step is not constant, write new time step**
14. End of program